

s-BLOCK ELEMENTS

1. Which of the following gives H_2O_2 on hydrolysis?
 (1) Na_2O (2) Na_2O_2
 (3) KO_2 (4) 2 & 3 both
2. Which of the following is coloured?
 (1) Li_2O (2) Na_2O_2 (3) KO_2 (4) Na_2O
3. Correct order of melting points & boiling point is—
 (M = Alkali metal)
 (1) $\text{MF} > \text{MBr} > \text{MI} > \text{MCl}$
 (2) $\text{MF} > \text{MCl} > \text{MBr} > \text{MI}$
 (3) $\text{MI} > \text{MBr} > \text{MCl} > \text{MF}$
 (4) $\text{MI} > \text{MBr} > \text{MF} > \text{MCl}$
4. Which of the following releases brown gas on heating?
 (1) NaNO_3 (2) LiNO_3 (3) KNO_3 (4) RbNO_3
5. Which of the following does not exist in solid form?
 (1) NaHCO_3 (2) $\text{Ba}(\text{HCO}_3)_2$
 (3) KHCO_3 (4) RbHCO_3
6. Which of the following on heating releases a gas which gives lime water test?
 (1) Li_2SO_4 (2) Na_2SO_4
 (3) K_2SO_4 (4) Rb_2SO_4
7. Which of the following forms chlorobridged dimer in vapour phase?
 (1) LiCl (2) BeCl_2 (3) BaCl_2 (4) MgCl_2
8. Tendency to form halide hydrates is maximum in—
 (1) MgCl_2 (2) CaCl_2 (3) SrCl_2 (4) BaCl_2
9. BeF_2 is prepared by thermal decomposition of
 (1) BeCO_3 (2) $(\text{NH}_4)_2\text{BeF}_4$
 (3) BeF_4^{2-} (4) BeSO_4
10. KHCO_3 cannot be prepared by solvay's process due to—
 (1) Unstability
 (2) High lattice energy
 (3) High solubility in water
 (4) High polarising power of K^+
11. On exposure to air, how many water molecules are lost by washing soda ($\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$) :—
 (1) 10 (2) 9 (3) 1 (4) 5
12. In Downs cell KCl is added in NaCl to—
 (1) Lowers its melting point
 (2) Dissolve more of NaCl
 (3) Lower conductivity
 (4) Increase the dissociation
13. Sodium carbonate solution in H_2O is alkaline due to—
 (1) Hydrolysis of Na^+
 (2) Hydrolysis of CO_3^{2-}
 (3) Hydrolysis of both Na^+ and CO_3^{2-}
 (4) None
14. In the Manufacturing of NaOH , by product obtained is—
 (1) O_2 (2) Cl_2 (3) Na_2CO_3 (4) NaCl
15. Setting of plaster of paris involves—
 (1) Oxidation with atmospheric oxygen
 (2) Combination with atmospheric CO_2
 (3) Dehydration
 (4) Hydration to yield another Hydrate
16. A metal 'M' readily form its sulphate MSO_4 . Which is water soluble. It form oxide MO which becomes inert, on heating. It forms insoluble hydroxide which is soluble in NaOH . The metal 'M' is—
 (1) Mg (2) Ba (3) Ca (4) Be
17. In curing cement plaster. Water is sprinkled from time to time this helps in—
 (1) Hydrating sand and gravel mixed with cement
 (2) Converting sand into silicate
 (3) Developing interlocked needle like crystals of hydrated silicates
 (4) Keeping it cool.

- 18.** A sodium salt on treatment with the MgCl_2 gives, white ppt only on heating, the anion of sodium salt is—
- (1) HCO_3^- (2) CO_3^{2-}
(3) NO_3^- (4) SO_4^{2-}
- 19.** Gypsum on heating at $120 - 130^\circ\text{C}$ gives—
- (1) Hemihydrate (2) Monohydrate
(3) Dihydrates (4) Anhydrous salt
- 20.** Which compound will liberate oxygen when react with H_2O —
- (1) Na_2O_2 (2) KO_2 (3) Na_2O (4) Cs_2O_2
- 21.** The compound formed upon combustion of sodium metal in excess of air—
- (1) Na_2O_2 (2) NaO_2 (3) NaOH (4) Na_2SO_4